GENERAL

G1. THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANTS' DRAWINGS, SPECIFICATIONS AND WITH SUCH OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. ANY DISCREPANCY SHALL BE REFERRED TO THE SUPERINTENDENT FOR DECISION BEFORE PROCEEDING WITH THE WORK.

G2. DIMENSIONS SHALL NOT BE OBTAINED BY SCALING THE STRUCTURAL DRAWINGS.

G3. SETTING OUT DIMENSIONS SHOWN ON THE DRAWINGS SHALL BE VERIFIED BY THE BUILDER.

G4. DURING CONSTRUCTION THE STRUCTURE SHALL BE MAINTAINED IN A STABLE CONDITION AND NO PART SHALL BE OVERSTRESSED. TEMPORARY BRACING SHALL BE PROVIDED BY THE BUILDER AS REQUIRED, EXCEPT FOR TEMPORARY WORKS DETAILED IN THESE DRAWINGS.

G5. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF THE SAA CODES AND THE BY-LAWS AND ORDINANCES OF THE RELEVANT BUILDING AUTHORITIES.

G6. REFER TO ARCHITECTURAL DRAWINGS FOR BRICK WALL THICKNESSES WHERE NOT MENTIONED ON THESE DRAWINGS AND FOR FALLS IN SLABS, EXTRA PACKING, WATERPROOFING MEMBRANES, CONTRACTION JOINT FILLING MATERIALS AND ALL OTHER ARCHITECTURAL FEATURES SUCH AS DRIP GROOVES, POUR BREAKS IN OFF-FORM CONCRETE, FILLETS AND THE LIKE.

G7. UNLESS NOTED OTHERWISE, ALL LEVELS ARE IN METRES AND ALL DIMENSIONS ARE IN MILLIMETRES.

G8. THE STRUCTURAL COMPONENTS DETAILED ON THESE DRAWINGS HAVE BEEN DESIGNED IN ACCORDANCE WITH THE RELEVANT SAA CODES AND LOCAL GOVERNMENT ORDINANCES FOR THE FOLLOWING LOADINGS :-

CLASSIFICATION	LIVE LOAD (KPa)
BALCONY	2.0
RESIDENCE	1.5
CARPARK	2.5

G9. THE RELEVANT PROVISIONS OF AS.1170.4 HAVE BEEN APPLIED FOR EARTHQUAKE DESIGN.

G10. THE RELEVANT PROVISIONS OF AS 1170.2 HAVE BEEN APPLIED FOR WIND LOADS WITH THE FOLLOWING PARAMETERS.
REGION A2
BASIC WIND SPEED Vu = 45m/s
TERRAIN CATEGORY 3

REINFORCEMENT

R1. REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY. IT IS NOT NECESSARILY SHOWN IN TRUE PROJECTION.

R2. SPLICES IN THE REINFORCEMENT SHALL BE MADE ONLY IN THE POSITIONS SHOWN. THE WRITTEN APPROVAL OF THE ENGINEER SHALL BE OBTAINED FOR ANY OTHER SPLICES. WHERE THE LAP LENGTH IS NOT SHOWN IT SHALL BE SUFFICIENT TO DEVELOP THE FULL STRENGTH OF THE REINFORCEMENT.

R3. WELDING OF REINFORCEMENT WILL NOT BE PERMITTED UNLESS SHOWN ON THE STRUCTURAL DRAWINGS OR APPROVED IN WRITING BY THE ENGINEER

R4. REINFORCEMENT SYMBOLS:-

N - DENOTES GRADE 500 NORMAL DUCTILITY DEFORMED BARS (D500N) TO AS/NZS 4671

R - DENOTES GRADE 250 NORMAL DUCTILITY ROUND BARS (R250N) TO AS/NZS 4671

SL- DENOTES GRADE 500 LOW DUCTILITY DEFORMED BAR SQUARE WELDED MESH (D500SL) TO AS/NZS 4671

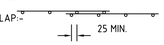
RL- DENOTES GRADE 500 LOW DUCTILITY DEFORMED BAR RECTANGULAR WELDED MESH (D500RL) TO AS/NZS 4671

THE FIGURES FOLLOWING WELDED MESH SYMBOLS `SL' OR `RL' ARE THE REFERENCE NUMBER FOR THE MESH TYPE IN ACCORDANCE WITH AS/NZS 4671

NUMBER OF BARS IN GROUP — BAR GRADE AND TYPE
4N20-200
NOMINAL BAR SIZE IN mm — SPACING IN mm

R5. ALL WELDED MESH SHALL COMPLY WITH AS/NZ 4671 AND SHALL BE SUPPLIED AS FLAT SHEETS.

TYPICAL WELDED MESH LAP:-



WELDED MESH SHALL BE LAPPED 2 TRANSVERSE WIRES PLUS 50mm. BUNDLED BARS SHALL BE TIED TOGETHER AT 30 BAR DIAMETER CENTRES WITH 3 WRAPS OF WIRE.

R6. PLACE SUFFICIENT BAR CHAIRS UNDER BOTTOM REINFORCING RODS AND TOP CROSS RODS IN SLABS TO ALLOW THEM TO BE SUPPORTED IN THEIR CORRECT POSITIONS DURING CONCRETING (NOT GREATER THAN 900mm CENTRES BOTH WAYS FOR BARS, 750mm FOR FABRIC).

R7. BAR CHAIR TYPE TO BE USED AS FOLLOWS:

EXPOSURE CLASSIFICATION	BAR CHAIR TYPE
A1, A2 & B1	PLASTIC-TIPPED WIRE OR PLASTIC
B2 & C	PLASTIC ONLY

R8. REINFORCEMENT LAYERS DENOTED THUS; (UNLESS NOTED OTHERWISE)-
TT - DENOTES TOP BARS LAID LAST IN TOP
T - DENOTES TOP BARS LAID FIRST IN TOP
B - DENOTES BOTTOM BARS LAID SECOND IN BTM
BB - DENOTES BOTTOM BARS LAID FIRST IN BTM

R9. WHERE TRANSVERSE TIE BARS ARE NOT SHOWN PROVIDE N12-300 SPLICED WHERE NECESSARY AND LAP WITH MAIN BARS 450mm, UNLESS NOTED OTHERWISE.

R10. MINIMUM LAP LENGTHS UNLESS NOTED OTHERWISE SHALL BE
500 FOR SIZE 12 BARS
600 FOR SIZE 16 BARS
900 FOR SIZE 20 BARS
1000 FOR SIZE 24 BARS
1200 FOR SIZE 28 BARS
1500 FOR SIZE 32 BARS
1800 FOR SIZE 36 BARS

BRICKWORK & BLOCKWORK

B1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS.3700.

B2. STRENGTHS OF BRICKS AND TYPE OF MORTAR SHALL BE AS FOLLOWS:-

ELEMENT	MATERIAL	CHARACTERISTIC UNCONFINED COMPRESSIVE STRENGTH (f'uc)	MORTAR TYPE C : L : S
BRICKWORK BLOCKWORK	BURNT CLAY CONCRETE	20 MPa (MIN.) 12 MPa (MIN.)	1 : 1 : 6 1 : ¾ : 3

B3. MAXIMUM DEPTH OF RAKING OF THE MORTAR JOINTS = 3mm. (UNO)

B4. BRICKWORK SUPPORTING SLABS AND BEAMS SHALL HAVE GALVANISED SLIDING JOINT STRIPS COMPRISING OF 2 GALVANISED STRIPS WITH GRAPHITE GREASE BETWEEN THEM (`SWAN' JOINT STRIPS OR EQUIVALENT).

B5. ALL BRICKWORK SUPPORTING OR SUPPORTED BY CONCRETE FLOORS SHALL BE PROVIDED WITH VERTICAL JOINTS TO MATCH ANY CONTROL JOINTS IN THE CONCRETE.

B6. NO MASONRY WALLS ARE TO BE ERECTED ON SUSPENDED SLABS OR BEAMS UNTIL ALL PROPPING HAS BEEN REMOVED.

B7. BUILDER TO PROVIDE TEMPORARY PROPPING TO ALL WALLS WHERE REQUIRED FOR STABILITY DURING CONSTRUCTION.

B8. OPENINGS FOR CLEAN-OUTS AND INSPECTIONS ARE TO BE PROVIDED AT THE BOTTOM COURSE OF EACH WALL FOR ALL CORES CONTAINING REINFORCEMENT. MORTAR PROTRUSIONS FROM ALL CORES ARE TO BE CLEANED OUT PRIOR TO PLACING REINFORCEMENT.

B9. NO CHASES OR RECESSES ARE PERMITTED IN MASONRY WITHOUT THE APPROVAL OF THE ENGINEER.

B10. BACKFILL TO RETAINING WALLS TO BE FREE DRAINING GRANULAR MATERIAL. PROVIDE SUBSOIL DRAIN OR WEEP HOLES.

B11. GALVANISED MASONRY COURSE REINFORCEMENT EQUAL TO SMORGON A.R.C. PRODUCT SHALL BE PROVIDED AT VERTICAL SPACINGS OF EVERY 6TH COURSE TO EACH LEAF AS FOLLOWS:
110mm MRH3650
140mm MRH3690
190mm MRH36140
230mm MRH36140
COURSE REINFORCEMENT SHALL BE LAPPED 300mm. ONE ADDITIONAL RUN SHALL BE PROVIDED UNDER & OVER ALL WINDOW & DOOR OPENINGS BONDED 300mm PAST THE OPENING.

B12. PROVIDE CONTROL JOINTS V.J. AT A MAXIMUM 4 METRE SPACINGS TYPICALLY & 4 METRES FROM CORNERS UNLESS NOTED OTHERWISE IN THE SPECIFICATION OR ON THESE DRAWINGS.

B13. DURABILITY

	EXTERNAL MASONRY	INTERNAL MASONRY (ABOVE DAMP PROOF COURSE AND INCLUDES INTERNAL LEAF OF CAVITY WALL)
EXPOSURE ENVIRONMENT	MODERATE	INTERIOR
MINIMUM SALT ATTACK RESISTANCE GRADE OF MASONRY UNITS (AS/NZS 4456.10)	`EXPOSURE'	`PROTECTED'
MINIMUM CLASSIFICATION OF MORTAR	M3	M2
MINIMUM DURABILITY CLASSIFICATION OF BUILT IN COMPONENTS (REINF., WALL TIES, BOLTS, MASONRY ANCHORS ETC.)	R3	R1
MINIMUM COVER TO REINFORCEMENT	15mm	5mm
WALL TIES (IN ACCORDANCE WITH AS2699.1) AND REINFORCEMENT	TO AS2699.1	TO AS2699.1
LINTELS (ETC)	GALVANISED (DUPLEX COATED) TO TABLE 3 - AS 2699.3	HOT-DIP GALVANISED TO AS 4680 (300g/m2)

FOUNDATIONS

F1. FOOTINGS HAVE BEEN DESIGNED FOR AN ALLOWABLE INTENSITY OR BEARING PRESSURE OF:

ELEMENT	STRATA	ALLOWABLE BEARING PRESSURE kPa
	HARD CLAY	250

NOTE
GEOTECHNICAL ENGINEER TO CONFIRM FOUNDING CAPACITY OF FOUNDATIONS PRIOR TO CONCRETE BEING PLACED.
BUILDER TO ALLOW FOR GEOTECHNICAL ENGINEERING COSTS.

F2. FOOTINGS SHALL BE LOCATED CENTRALLY UNDER WALLS AND COLUMNS UNLESS NOTED OTHERWISE.

F3. SITE IS ASSUMED TO BE M SITE AS PER AS2870. NEED TO BE CONFIRMED BY GEOTECHNICAL ENGINEER.

F4. EXCAVATE FOR FOOTINGS TO THE NOMINATED SIZE AND DEPTH. FOUNDING LEVELS ARE PROVISIONAL SUBJECT TO ACTUAL SIZE CONDITIONS AND APPROVAL BY THE GEOTECHNICAL ENGINEER.

F5. DO NOT BACKFILL RETAINING WALLS (OTHER THAN CANTILEVER WALLS) UNTIL FLOOR CONSTRUCTION AT TOP AND BOTTOM IS COMPLETED (UNO). ENSURE FREE DRAINING BACKFILL AND DRAINAGE IS IN PLACE.

F6. FOOTINGS TO BE CONSTRUCTED AND BACKFILLED AS SOON AS POSSIBLE FOLLOWING EXCAVATION TO AVOID SOFTENING OR DRYING OUT BY EXPOSURE.

F7. THESE DRAWINGS ARE TO BE READ IN CONJUNCTION WITH GEOTECHNICAL INVESTIGATION REPORT BY :

DECLARATION

A SITE-SPECIFIC GEOTECHNICAL REPORT SHALL BE OBTAINED AND PROVIDED TO THIS OFFICE FOR ENGINEERING REVIEW PRIOR TO COMMENCEMENT OF CONSTRUCTION. IN THE ABSENCE OF ADEQUATE GEOTECHNICAL INFORMATION, THIS OFFICE RESERVES THE RIGHT TO DECLINE ANY INSPECTION OR CERTIFICATION ASSOCIATED WITH THE WORKS.

CONCRETE

C1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH ACSE CONCRETE SPECIFICATION AND AS.3600 WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.

C2. CONCRETE QUALITY (UNO ON PLANS):

ELEMENT	f'c MPa (28 DAYS)	SLUMP	MAX. AGG. SIZE	CEMENT TYPE	SHRINKAGE
FOOTING	25	80	20	GB	700
SUSPENDED SLAB	40	80	20	GB	700
SLAB ON GROUND	32	80	20	GB	700
GROUT IN BLOCKWORK	25	225	10	GB	-
COLUMNS	32	80	20	GB	-

PROJECT CONTROL TESTING SHALL BE CARRIED OUT IN ACCORDANCE WITH AS 1379.

C3. NO ADMIXTURES SHALL BE USED IN CONCRETE UNLESS APPROVED IN WRITING.

C4. CLEAR CONCRETE COVER TO ALL REINFORCEMENT FOR DURABILITY SHALL BE AS FOLLOWS, UNLESS SHOWN OTHERWISE :-

EXPOSURE CLASSIFICATION TO AS 3600	CONCRETE GRADE (MIN.)	CAST AGAINST GROUND	CAST IN FORMS AND EXPOSED	CAST IN FORMS NOT EXPOSED
A1 & A2	25	50mm	30mm	20mm (A1)
B1	32	60mm	40mm	
B2	40	65mm	45mm	

NOTE: (i) COVER REQUIREMENTS MAY NEED TO BE INCREASED TO SUIT FIRE RATING.
(ii) EXPOSURE CLASSIFICATION SHALL BE AS INDICATED BELOW OR ON THE DRAWINGS.
(iii) THE SURFACES OF MEMBERS PROTECTED FROM THE GROUND BY A DAMP-PROOF MEMBRANE HAVE EXPOSURE CLASSIFICATION A1, UNLESS NOTED OTHERWISE.

C5. FOR CHAMFERS, DRIP GROOVES, REGLETS, etc., REFER TO ARCH'S. DETAILS, MAINTAIN COVER TO REINFORCEMENT AT THESE DETAILS

C6. SIZES OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF APPLIED FINISHES.

C7. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE ENGINEER.

C8. BEAM DEPTHS ARE WRITTEN FIRST AND INCLUDE SLAB THICKNESS, IF ANY.

C9. NO HOLES OR CHASES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS WITHOUT THE PRIOR APPROVAL OF THE ENGINEER.

C10. PIPES OR CONDUITS SHALL NOT BE PLACED WITHIN THE CONCRETE COVER TO REINFORCEMENT WITHOUT THE APPROVAL OF THE ENGINEER. THE CONCRETE COVER TO EMBEDDED PIPES OR CONDUITS SHALL BE A MINIMUM OF 20mm.

C11. ALL THE REQUIREMENTS OF THE CONCRETE SPECIFICATION SHALL APPLY TO THE FORMWORK, REINFORCEMENT AND CONCRETE.

C12. THE CONCRETE WORK SHOWN ON THESE DRAWINGS HAS BEEN DESIGNED FOR THE FOLLOWING EXPOSURE CLASSIFICATION FOR DURABILITY IN ACCORDANCE WITH AS.3600.

ELEMENT	EXPOSURE CLASSIFICATION
INTERNAL EXTERNAL	A1 B1

C13. THE FINISHED CONCRETE SHALL BE A DENSE HOMOGENEOUS MASS, COMPLETELY FILLING THE FORMWORK THOROUGHLY EMBEDDING THE REINFORCEMENT AND FREE OF STONE POCKETS. ALL CONCRETE INCLUDING SLABS ON GROUND AND FOOTINGS SHALL BE COMPACTED WITH MECHANICAL VIBRATORS.

C14. CURING OF ALL CONCRETE IS TO BE ACHIEVED BY KEEPING SURFACES CONTINUOUSLY WET FOR A PERIOD OF 3 DAYS, AND PREVENTION OF LOSS OF MOISTURE FOR A TOTAL OF 7 DAYS FOLLOWED BY A GRADUAL DRYING OUT. APPROVED SPRAYED ON CURING COMPOUNDS MAY BE USED WHERE NO FLOOR FINISHES ARE PROPOSED. POLYTHENE SHEETING OR WET HESSIAN MAY BE USED IF PROTECTED FROM WIND AND TRAFFIC.

C15. CONSTRUCTION SUPPORT PROPPING IS TO BE LEFT IN PLACE WHERE NEEDED TO AVOID OVERSTRESSING THE STRUCTURE DUE TO CONSTRUCTION LOADING. NO BRICKWORK OR PARTITION WALLS ARE TO BE CONSTRUCTED ON SUSPENDED LEVELS UNTIL PROPPING IS REMOVED AND THE SLAB HAS ABSORBED ITS DEAD LOAD DEFLECTION.

C16. THE ENGINEER SHALL BE GIVEN 48 HOURS NOTICE FOR REINFORCEMENT INSPECTION AND CONCRETE SHALL NOT BE DELIVERED UNTIL FINAL APPROVAL IS OBTAINED.

C17. CONDUITS, PIPES, etc., SHALL ONLY BE LOCATED IN THE MIDDLE ONE THIRD OF SLAB DEPTH AND SPACED AT NOT LESS THAN 3 DIAMETERS.

C18. PROVIDE UPWARD CAMBER TO FORMWORK OF CANTILEVERS OF L/120, WHERE `L' IS THE SHORTEST PROJECTION BEYOND COLUMN OR WALL FACE, AND TO FORMWORK OF SLABS WHERE NOTED ON PLAN. MAINTAIN THE SLAB AND BEAM DEPTHS AS SHOWN.

TIMBER

T1. ALL TIMBER DESIGN, CONSTRUCTION, WORKMANSHIP AND MATERIAL SHALL BE IN STRICT ACCORDANCE WITH AS 1720.1 SAA TIMBER STRUCTURES CODE.

T2. AS1684 - NATIONAL TIMBER FRAMING CODE SHALL BE APPLIED WHERE APPROPRIATE TO DOMESTIC CONSTRUCTION

T3. ALL SOFTWOOD TIMBER SHALL BE MINIMUM F7 STRESS GRADE UNLESS NOTED OTHERWISE. ALL HARDWOOD TO BE MINIMUM GRADE F11 (UNO)

T4. UNLESS NOTED OTHERWISE ALL BOLTS IN TIMBER CONSTRUCTION TO BE COMMERCIAL BOLTS OF GRADE 4.6 SNUG TIGHT (M16-4.6/S) CONFORMING TO AS 1111 WITH WASHERS AS SPECIFIED.

T5. END AND EDGE DISTANCES FOR BOLTS WHERE NOT SPECIFIED SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF AS 1720.1.

T6. ALL BOLTS, NUTS & WASHERS TO BE HOT DIP GALVANISED, OR STAINLESS STEEL GRADE A4 (316), NOT ZINC OR CADMIUM PLATED.

T7. EXTERNAL TIMBER TO BE EITHER HARDWOOD DURABILITY CLASS II OR II TO AS1720.2 OR IMPREGNATED PINE GRADE F7, PRESSURE TESTED TO AS1604 AND RE-DRIED PRIOR TO USE. SUPPLEMENTARY TREATMENT SHALL BE APPLIED TO CUT SURFACES.

STEELWORK

S1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH ACSE STRUCTURAL STEELWORK SPECIFICATION AND AS.4100 AND AS.1554 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.

S2. UNLESS OTHERWISE NOTED, ALL STEEL SHALL BE IN ACCORDANCE WITH:-
AS/NZS.3679.1 GRADE FOR BHP-300 PLUS ROLLED SECTIONS AND MERCHANT BAR EXCEPT WHERE NOTED.
AS/NZS.3679.2 GRADE 300 FOR ALL WELDED SECTIONS (WB & WC)
AS.1163 GRADE C350 FOR RECTANGULAR AND SQUARE HOLLOW SECTIONS.
AS.1163 GRADE C350 FOR CIRCULAR HOLLOW SECTIONS GREATER THAN 88.9ø.
AS.1163 GRADE C250 FOR REMAINDER OF CIRCULAR HOLLOW SECTIONS.
AS.1397 COLD FORMED STEEL PURLINS AND GIRTS.
AS.3678 GRADE 250 FOR ALL PLATE

S3. THE BUILDER SHALL PREPARE WORKSHOP DRAWINGS AND SHALL SUBMIT COPIES OF EACH DRAWING FOR REVIEW. ALLOW 7 DAYS FOR SHOP DRAWING REVIEW. FABRICATION SHALL NOT COMMENCE UNTIL REVIEW HAS BEEN COMPLETED. REVIEW DOES NOT INCLUDE DIMENSIONS.

S4. HIGH STRENGTH STRUCTURAL BOLTS AND WASHERS SHALL COMPLY WITH AS.1252. COMMERCIAL GRADE BOLTS, NUTS AND WASHERS SHALL COMPLY WITH AS.1111, AS.1112 AND AS.1237 RESPECTIVELY. HIGH STRENGTH FRICTION GRIP BOLTS, NUTS AND WASHERS SHALL BE INSTALLED IN ACCORDANCE WITH AS.4100 AND SHALL BE TIGHTENED TO THE CORRECT TENSION USING APPROVED LOAD INDICATING WASHERS. CONTACT SURFACES OF ALL HIGH STRENGTH FRICTION GRIP BOLTED CONNECTIONS SHALL BE LEFT UNPAINTED. UNLESS NOTED OTHERWISE, ALL BOLTS TO BE 20 DIAMETER HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 SNUG TIGHT (M20-8.8/S) (EXCEPT PURLIN & H.D BOLTS)
BOLTS - DESIGNATED BY THE NUMBER, DIAMETER, GRADE AND TIGHTENING PROCEDURE.
eg. 4M16-4.6/S MEANS 4M16 DIA. COMMERCIAL GRADE BOLTS SNUG TIGHT.
6M20-8.8TF MEANS 6M20 HIGH STRENGTH STRUCTURAL BOLTS FULLY TENSIONED IN A FRICTION JOINT.
6M24-8.8TB MEANS 6M24 HIGH STRENGTH STRUCTURAL BOLTS FULLY TENSIONED IN A BEARING JOINT. (SOME SLIP ALLOWED.)

ALL CONNECTIONS TO HAVE 2 BOLTS MIN. PER CONNECTION WITH GUSSET PLATES 10 THICK UNLESS NOTED OTHERWISE.

S5. ALL WELDING DESIGN AND FABRICATION SHALL CONFORM TO AS.1554. UNLESS OTHERWISE NOTED, ALL WELDS TO BE 6mm CONTINUOUS FILLET FROM E48XX ELECTRODES.
ALL STRUCTURAL WELDS SHALL BE CATEGORY SP UNLESS NOTED OTHERWISE.
GENERAL PURPOSE WELDS SHALL BE DENOTED THUS `GP'.
BUTT WELDS WHERE INDICATED IN THE DRAWINGS ARE TO BE COMPLETE PENETRATION BUTT WELDS AS DEFINED IN AS.1554.
WELDING SYMBOLS TO AS.1101 PART 3.

S6.(a) INTERNAL STEELWORK (U.N.O. IN SPECIFICATION/DRAWINGS)

STRUCTURAL STEELWORK WHICH WILL REMAIN FULLY ENCLOSED WITHIN BUILDING FOR THE DESIGN LIFE OF STRUCTURE SHALL RECEIVE A CORROSION PROTECTION SYSTEM APPROVED BY ARCHITECT/PROJECT SUPERINTENDENT, BUT NOT LESS THAN ON THE FOLLOWING:

IZS1 (INORGANIC ZINC SILICATE 75 um)
ILG300 (IN-LINE GALVANISED 42 um)
PGS300 (ELECTRO GALVANISED 42 um)

(b) EXTERNAL STEELWORK (EXTERIOR AND BUILT INTO EXTERNAL WALLS) - U.N.O. IN SPECIFICATION/DRAWINGS.

STRUCTURAL STEELWORK SHALL RECEIVE A CORROSION PROTECTION SYSTEM IN ACCORDANCE WITH AS/NZS2312 AND FOLLOWING PARAMETERS:

DURABILITY - YEARS TO FIRST MAINTENANCE: VERY LONG TERM (15-25 YEARS)
ATMOSPHERIC CORROSIVITY CATEGORY: D (HIGH)

WRITTEN STATEMENT OF COMPLIANCE WITH ABOVE SHALL BE SUBMITTED TO PROJECT SUPERINTENDENT PRIOR TO STEELWORK BEING DELIVERED TO SITE. AREAS WITH DAMAGED COATING SHALL BE REPAIRED SO THAT THE REPAIRED AREA HAS A DURABILITY EQUAL OR EXCEEDING THAT SPECIFIED.

S7. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER TO STEEL WHETHER OR NOT DETAILED IN THE DRAWINGS.

S8. ALL THE REQUIREMENTS OF THE STRUCTURAL STEEL SPECIFICATION SHALL APPLY. (U.N.O.)

S9. THE BUILDER IS TO BE PRESENT WHEN ALL HOLDING DOWN BOLTS ARE INSTALLED TO ENSURE THEY ARE NOT DISPLACED DURING CONCRETE PLACEMENT.

S10. THE BUILDER IS TO MAKE GOOD AND/OR REPAIR ALL DAMAGED SURFACES DURING PERFORMANCE OF THE WORK.

S11. UNLESS SHOWN ON THE DRAWINGS THE ROOF STRUCTURE HAS BEEN DESIGNED FOR NORMAL ROOF LOADS ONLY AND DOES NOT ALLOW FOR ANY EXTRANEOUS LOADS SUCH AS HOISTS, MONORAILS etc.

S12. PROVIDE SEAL PLATES TO ALL HOLLOW SECTIONS, WITH 'BREATHER' HOLES IF MEMBERS ARE TO BE HOT DIPPED GALVANISED.

S13. WELD INSPECTIONS SHALL BE CARRIED OUT AS FOLLOWS:-

VISUAL SCANNING - 100% OF ALL WELDS (U.N.O. ON DRAWINGS)

NON-DESTRUCTIVE EXAMINATION (AS 1554.1)

WELD CATEGORY	WELD APPLICATION	VISUAL EXAMINATION	ULTRASONIC OR MAGNETIC PARTICLE
GP	PURLIN CLEAT FILLET WELDS STIFFENER FILLET WELDS	10%	-
SP	BUTT WELDS	50%	5%
	ALL OTHER WELDS (UNO)	25%	-

ALL WELD INSPECTIONS SHALL BE CARRIED OUT BY AN INDEPENDENT NATA REGISTERED/CERTIFIED WELDING INSPECTOR. A REPORT APPROVING ALL WELDING SHALL BE FORWARDED TO THE ENGINEER FOR REVIEW PRIOR TO STRUCTURAL STEELWORK LEAVING THE FABRICATION FACILITIES.

DRAFTING ABBREVIATIONS

B.E.L.	BULK EXCAVATION LEVEL
B.I.F.B.	BITUMEN IMPREGNATED FIBRE BOARD
BLK.	BLOCKWORK
BWK.	BRICKWORK
B/W.	BOTH WAYS
CONC.	CONCRETE
C/C.	CENTRES
C/S.	COURSES
D.J.	DOWELLED JOINT
E.F.	EACH FACE
F.F.	FAR FACE
F.F.L.	FINISHED FLOOR LEVEL
F.G.L.	FINISHED GROUND LEVEL
F.R.L.	FIRE RESISTANCE LEVEL
K.J.	KEYED JOINT
MAX.	MAXIMUM
MIN.	MINIMUM
N.F.	NEAR FACE
N.S.O.P.	NOT SHOWN ON PLAN
N.T.S.	NOT TO SCALE
REINF.	REINFORCEMENT
TYP.	TYPICAL
U.N.O.	UNLESS NOTED OTHERWISE
V.B.M.	VAPOUR BARRIER MEMBRANE
V.J.	VERTICAL JOINT

NOTE

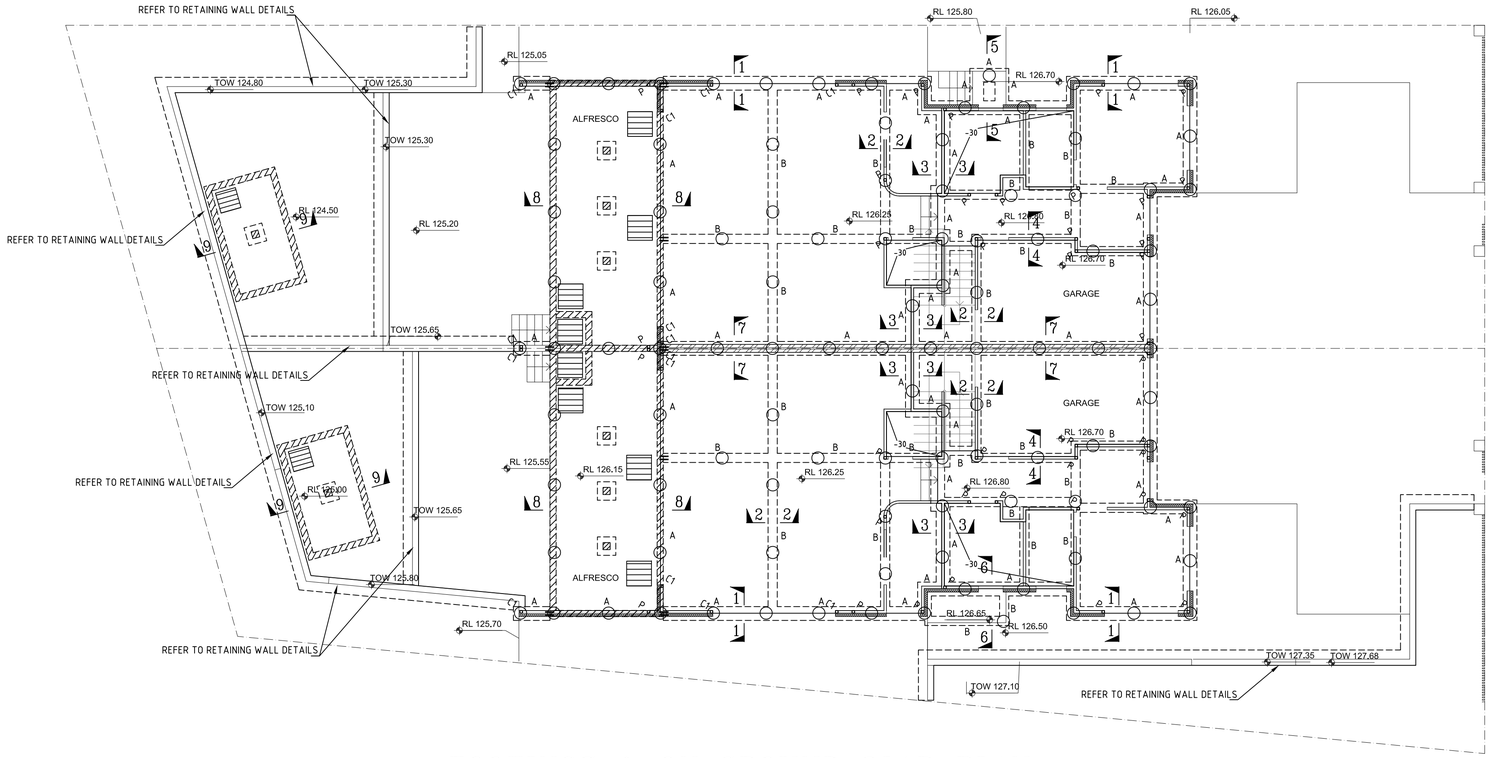
DETAILS AND SECTIONS ON THESE DRAWINGS ARE CROSS-REFERENCED BY THE FOLLOWING SYSTEM:-

SECTION OR DETAIL No. — 5
SHEET No., WHERE SECTION OR DETAIL IS DRAWN. — S03

client	date	issue	amendment	checked	date	issue	amendment	checked	project
	16.09.2026	A	ISSUED FOR CC	JZ					257 MARSDEN ROAD, CARLINGFORD

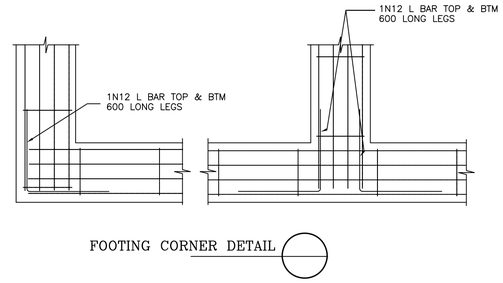
MBC ENGINEERING PTY LTD
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drawing title: STRUCTURAL NOTES
date 16/09/2026 designed: C.Z.(BE,MIE 3928680) drawing no. S01
job no. 2025466
scale 1:100;1:20 drafted: J.Z.

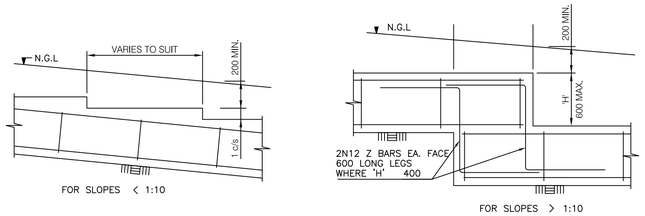


FOOTINGS & ON GROUND SLAB PLAN

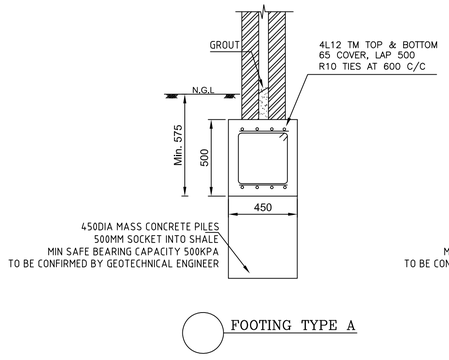
120mm ON GROUND SLAB; SL82 MESH TOP; 0.2mm POLYTHENE MEMBRANE ON 50mm LEVELING SAND,UNO
IF BACKFILL MORE THAN 500MM, 150MM SLAB ,USE SL82 MESH TOP&BOT,TYP.



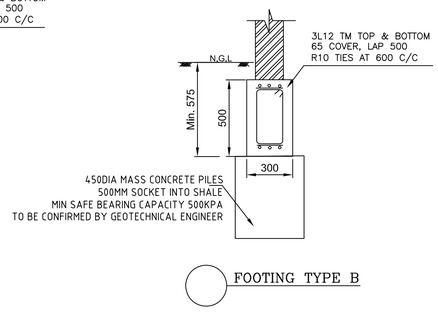
FOOTING CORNER DETAIL



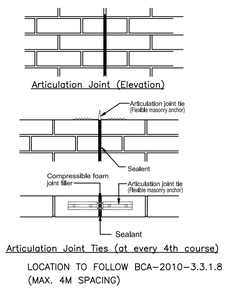
STRIP FOOTING STEP DETAILS



FOOTING TYPE A

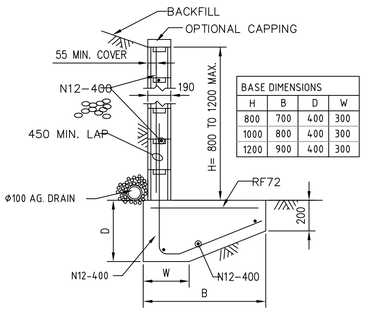


FOOTING TYPE B

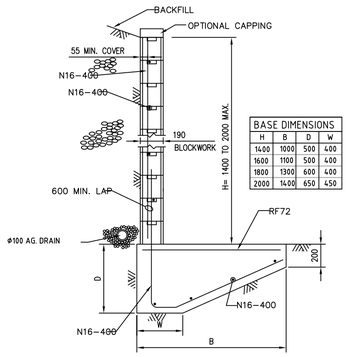


Articulation Joint Ties (at every 4th course)

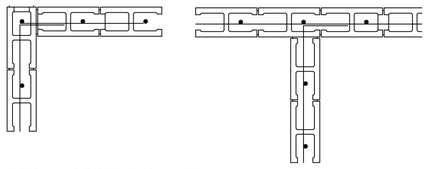
LOCATION TO FOLLOW BCA-2010-3.3.1.8
(MAX. 4M SPACING)



BASE DIMENSIONS			
H	B	D	W
800	700	400	300
1000	800	400	300
1200	900	400	300



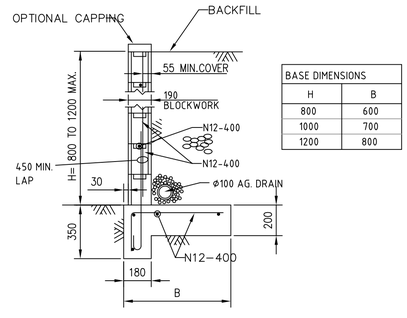
BASE DIMENSIONS			
H	B	D	W
1400	1000	500	400
1600	1100	500	400
1800	1300	600	400
2000	1400	650	450



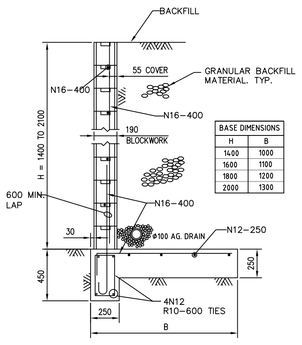
TYPICAL CORNER DETAIL

ALL BLOCKWORK
GROUT FILL CORES
GROUT STRENGTH F'c=25MPa
NOTE: PLACE CONSTRUCTION EVEN 6-8m.

TYPICAL BLOCK WALL DETAILS

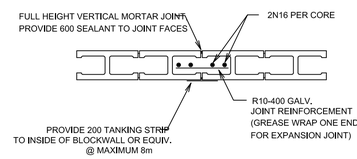


BASE DIMENSIONS	
H	B
800	600
1000	700
1200	800



BASE DIMENSIONS	
H	B
1400	1000
1600	1100
1800	1200
2000	1300

TYPICAL RETAINING WALL



TYPICAL CONSTRUCTION JOINT (CJ)

BLOCKWORK NOTES

1. ALL BLOCKS TO BE 140 DOUBLE "U" BLOCKS.
   PROVIDE CLEAN OUT HOLES TO BTM. COURSE.

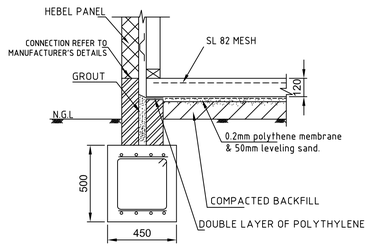
   FILL ALL CORES WITH CONCRETE:- 10mm AGG.
   25 MPa
   225 SLUMP
2. 50 MIN. COVER TO ALL BASE REINFORCEMENT.
3. BACKFILL TO BE COMPACTED CLEAN GRANULAR
   FREE DRAINING MATERAIL.

client	date	issue	amendment	checked	date	issue	amendment	checked	project
	16.09.2026	A	ISSUED FOR CC	JZ					257 MARSDEN ROAD, CARLINGFORD

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MOBILE:0432546227

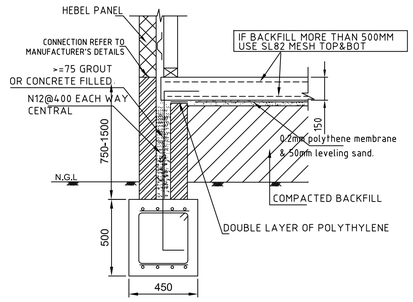
drawing title: GROUND FLOOR PLAN & DETAILS

date 16/09/2026
job no. 2025466
scale 1:100,1:20
designed: C.Z.(BE,MIE 3928680)
drafted: J.Z.
drawing no. S02



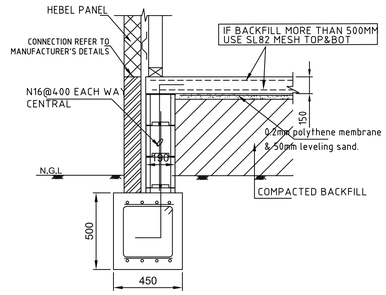
SECTION 1

NOTE:THE DEPTH OF BACKFILL LESS THAN 750



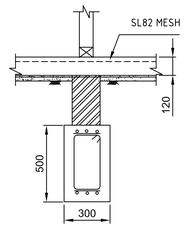
SECTION 1

BACKFILL MORE THAN 750MM

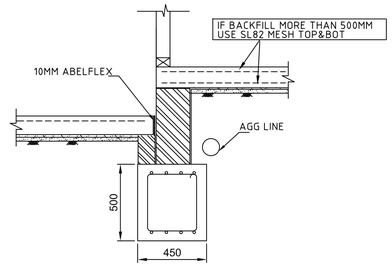


SECTION 1

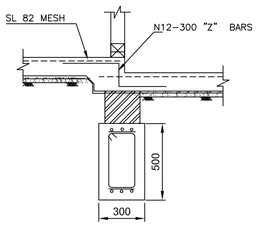
BACKFILL MORE THAN 1500MM



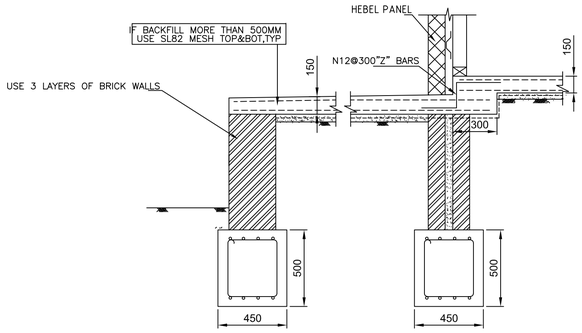
SECTION 2



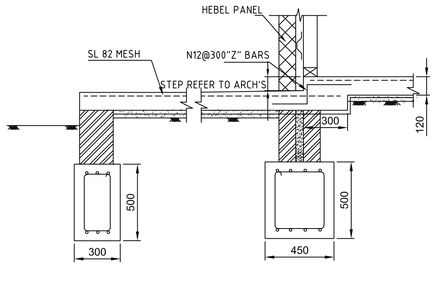
SECTION 3



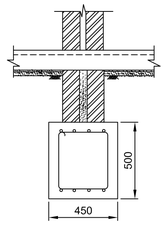
SECTION 4



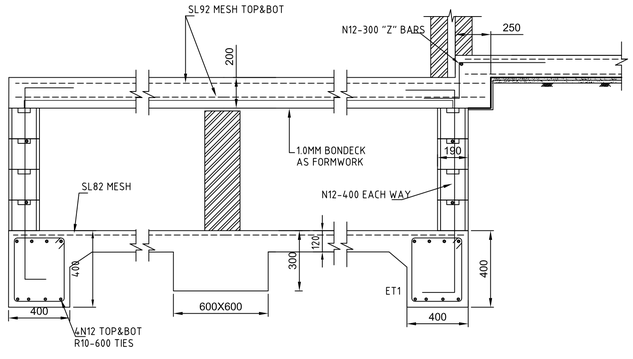
SECTION 5



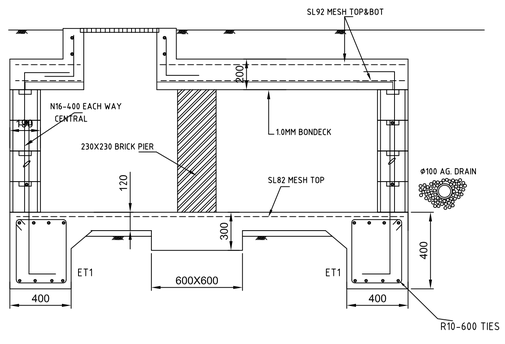
SECTION 6



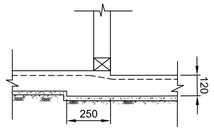
SECTION 7



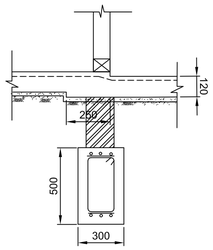
SECTION 8



SECTION 9



SECTION
WET AREA SETDOWN DETAIL



SECTION
WET AREA SETDOWN DETAIL

client	date	issue	amendment	checked	date	issue	amendment	checked	project
	16.09.2026	A	ISSUED FOR CC	JZ					257 MARSDEN ROAD, CARLINGFORD

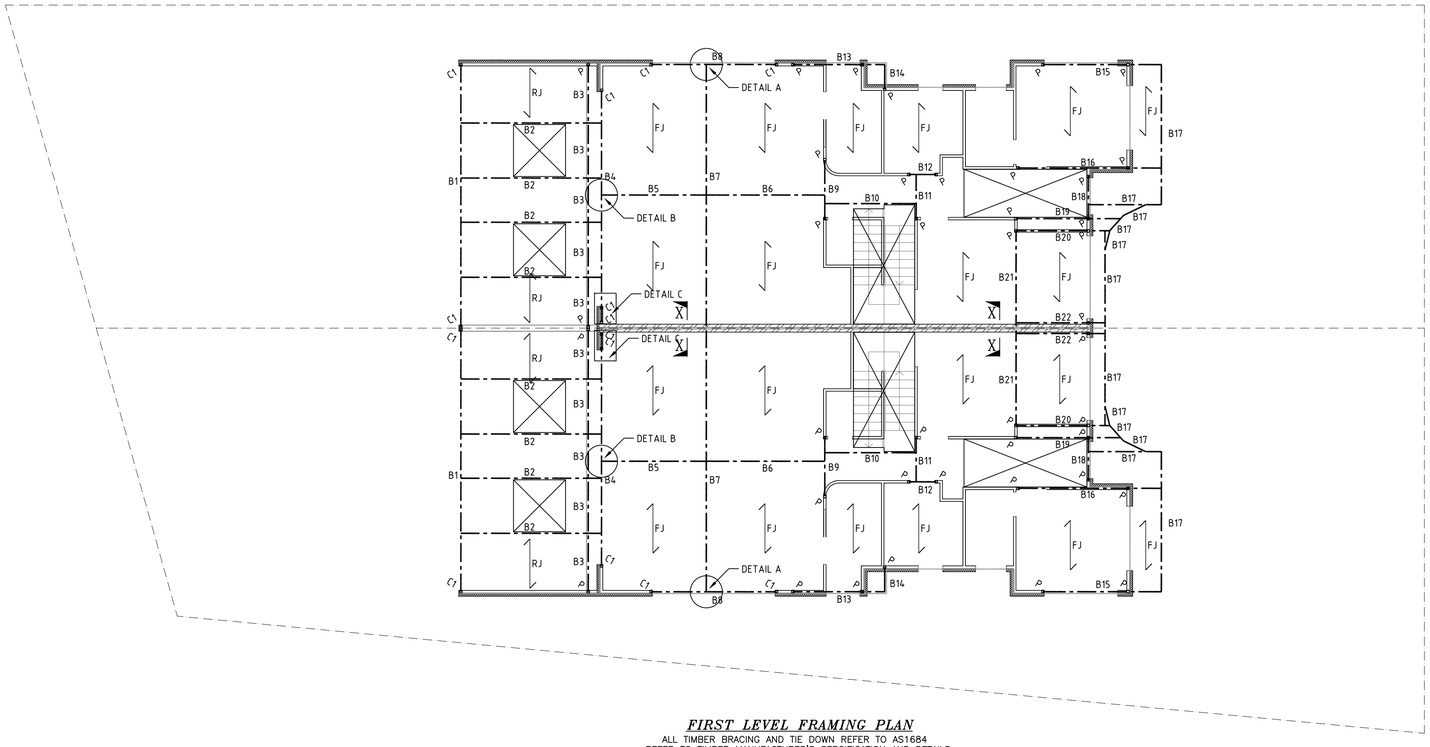
MBC ENGINEERING PTY LTD
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PO BOX 269
PARRAMATTA NSW 2124
MBCCONSULTING@LIVE.COM
MOBILE:0432546227

drawing title: GROUND FLOOR PLAN & DETAILS

date 16/09/2026 | designed: C.Z.(BE,MIE 3928680) | drawing no. S03

job no. 2025466 | drafted: J.Z.

scale 1:100,1:20



FIRST LEVEL FRAMING PLAN

ALL TIMBER BRACING AND TIE DOWN REFER TO AS1684
REFER TO TIMBER MANUFACTURER'S SEPCIFICATION AND DETAILS
HEBEL FLOOR IS ALLOWED FOR FIRST FLOOR SLAB

BEAM SCHEDULE

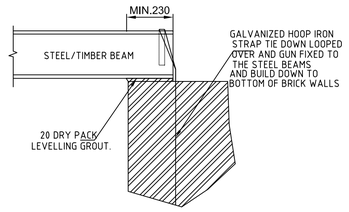
MARK No.	SIZE
B1	300PFC
B2	2/300X42 SMARTLVL
B3	2/240X42 SMARTLVL
B4	380PFC
B5	3/300X58 SMARTLVL
B6	3/300X58 SMARTLVL
B7	310UC 137
B8	300PFC
B9	2/300X42 SMARTLVL
B10	300X42 SMARTLVL
B11	300X42 SMARTLVL
B12	300X42 SMARTLVL
B13	2/240X42 SMARTLVL
B14	2/240X42 SMARTLVL
B15	2/240X42 SMARTLVL
B16	2/240X42 SMARTLVL
B17	2/240x42 SMARTLVL
B18	2/200X42 SMARTLVL
B19	2/200X42 SMARTLVL
B20	2/200X42 SMARTLVL
B21	2/240X42 SMARTLVL
B22	2/200X42 SMARTLVL
P	90X90 HW POST
C1	89X89X5.0 SHS
FJ	SMARTJOIST SJ30051@ 450 MAX. SPAN 4.4M

ALL EXTERNAL STEEL BEAM TO BE H.D.GALV.
ALL EXTERNAL TIMBER BEAM TO BE TREATED AS H3

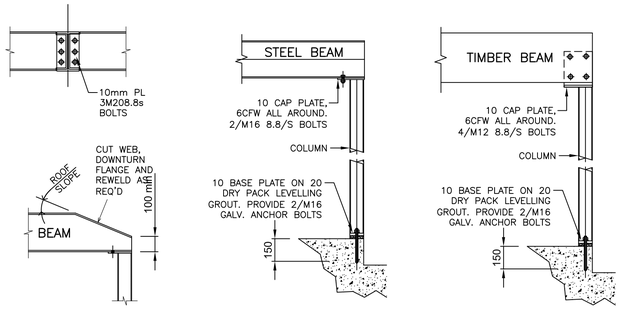
TIMBER STUD SIZE FOR WALLS	
H<3M	90X45 MGP10@ 450 C/C
3M<H<4.2M	90X58 SMARTLVL @ 300 C/C
5M>H>4.2M	130X58 SMARTLVL @ 300 C/C
6M>H>5M	150X58 SMARTLVL @ 300 C/C

PROVIDE DOUBLE STUD AT EACH SIDE OF WINDOW

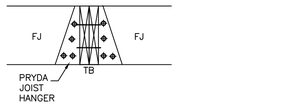
LINTEL SCHEDULE		
SPAN	INTERNAL SKIN	EXTERNAL SKIN
0~1000	2/90X45 MGP10	85X8 FLAT BAR
1001~2000	2/150X42 SMARTLVL	100X100X6 GALINTEL
2001~2500	2/200X42 SMARTLVL	150X100X6 GALINTEL
2501~3000	2/240X42 SMARTLVL	150X100X10 GALINTEL



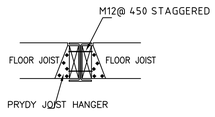
BRICK/BEAM CONNECTION DETAILS



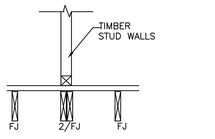
TYPICAL COLUMN/BEAM DETAILS



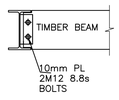
TYPICAL FLOOR JOIST TO TIMBER BEAM CONNECTION



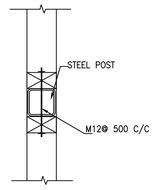
TYPICAL FLOOR JOIST TO STEEL BEAM CONNECTION



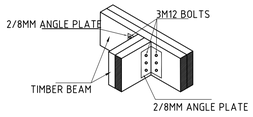
TYPICAL SECTION THROUGH LOAD BEARING WALLS



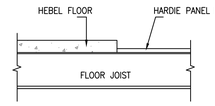
TYPICAL TIMBER BEAM TO STEEL BEAM CONNECTION



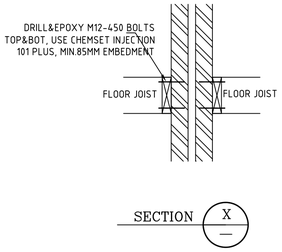
TYPICAL STEEL POST TO FRAMING WALL CONNECTION

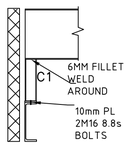


TIMBER BEAM CONNECTION

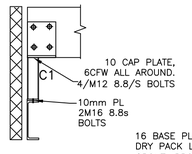


TYPICAL WET AREA SETDOWN DETAIL

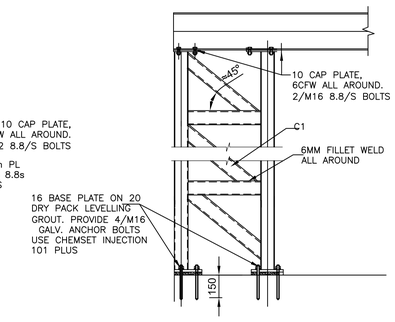




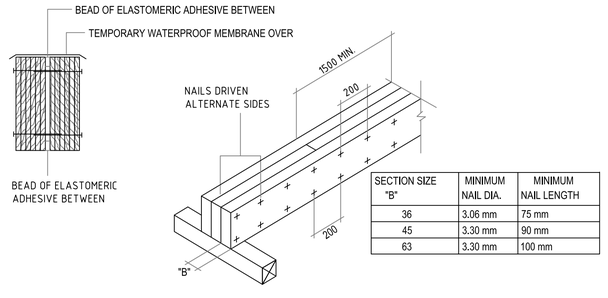
DETAIL A



DETAIL B

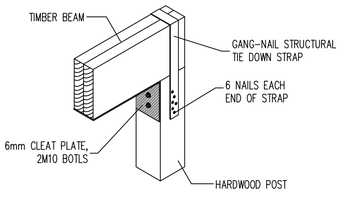


DETAIL C

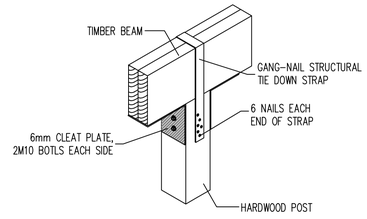


SECTION SIZE "B"	MINIMUM NAIL DIA.	MINIMUM NAIL LENGTH
36	3.06 mm	75 mm
45	3.30 mm	90 mm
63	3.30 mm	100 mm

VERTICAL LAMINATION DETAIL



TIMBER BEAM TO POST CONNECTION

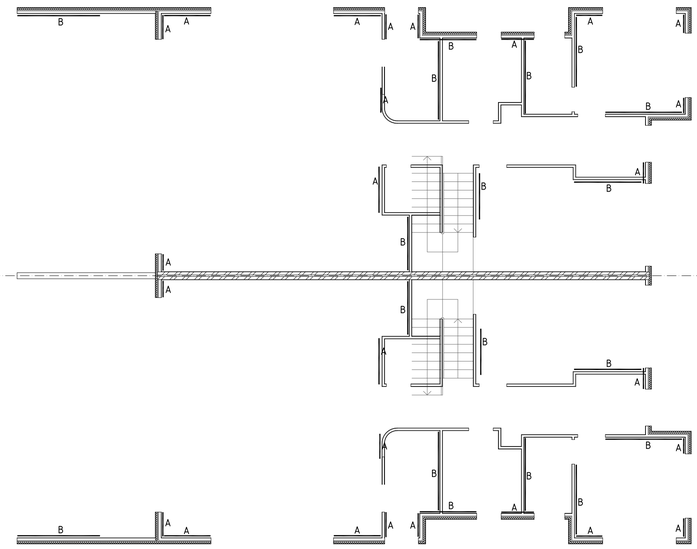


TIMBER BEAM TO POST CONNECTION

client	date	issue	amendment	checked	date	issue	amendment	checked	project
	16.09.2026	A	ISSUED FOR CC	JZ					257 MARSDEN ROAD, CARLINGFORD

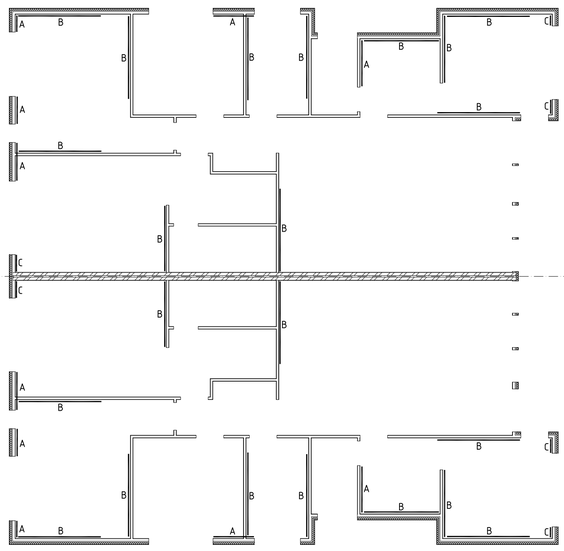
MBC ENGINEERING PTY LTD
BUILDING CIVIL & STRUCTURAL ENGINEERS
PO BOX 269
PARRAMATTA NSW 2124
MBCCONSULTING@LIVE.COM
MOBILE:0432546227

drawing title: FIRST FLOOR PLAN & DETAILS
date 16/09/2026
job no. 2025466
scale 1:100;1:20
designed: C.Z.(BE,MIE 3928680)
drafted: J.Z.
drawing no. S04



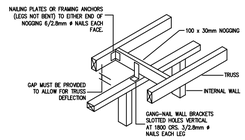
GROUND FLOOR BRACING LAYOUT

A– PLYWOOD BRACING REFER TO DETAIL
B– WALL STRAP BRACING REFER TO DETAIL
C– HARDBOARD BRACING REFER TO DETAIL

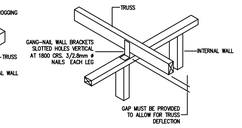


FIRST FLOOR BRACING DETAIL

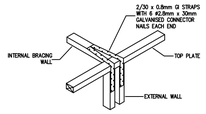
A– PLYWOOD BRACING REFER TO DETAIL
B– WALL STRAP BRACING REFER TO DETAIL
C– HARDBOARD BRACING REFER TO DETAIL



NON LOAD BEARING WALL PARALLEL TO TRUSSES

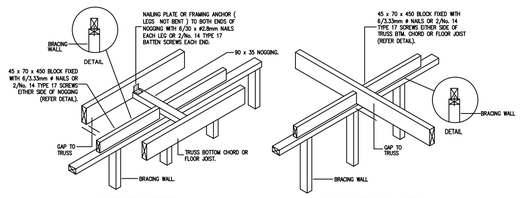


NON LOAD BEARING WALL PERPENDICULAR TO TRUSSES



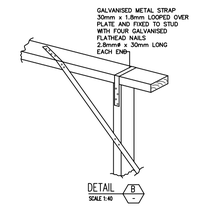
INTERNAL BRACING WALL TO EXTERNAL WALL CONNECTION

NOTE :
ALL BRACING TO COMPLY WITH RELEVANT SECTIONS OF AS1684 RESIDENTIAL TIMBER FRAMED CONSTRUCTION. FOR FURTHER INFORMATION REFER TO CLAUSE 8.3.8 OF PART 2 AS1684.2-1999



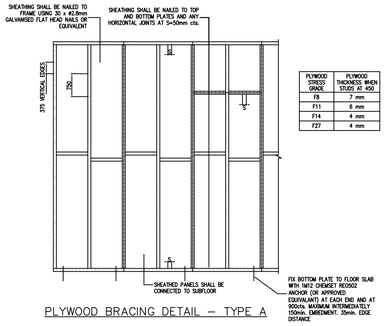
PARALLEL TO TRUSS PERPENDICULAR TO TRUSS

TIMBER BRACING WALL TOP CONNECTION DETAIL



DETAIL B
SCALE 1:40

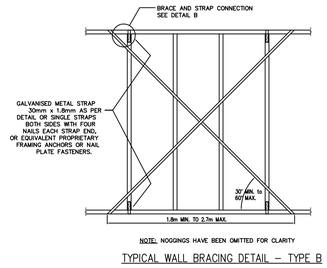
METAL TENSION STRAP BRACING
CORROSION PROTECTED FLAT METAL TENSION STRAPPING FIXED WITH TWO GALVANISED FLATHEAD NAILS 3.15mm# x 30mm LONG TO EACH END STUD AND THE FACE OF THE TOP AND BOTTOM PLATE AND FOUR GALVANISED FLATHEAD NAILS 3.15mm# x 30mm LONG TO THE STRAP RETURN OVER THE TOP PLATE AND UNDER THE BOTTOM PLATE.



PLYWOOD STRESS GRADE	PLYWOOD THICKNESS WHEN STUDS AT 450
F8	7 mm
F11	6 mm
F14	4 mm
F27	4 mm

PLYWOOD BRACING DETAIL – TYPE A

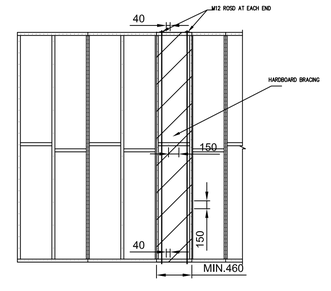
- PLYWOOD BRACING IN ACCORDANCE WITH TABLE 8.18(g) OF AS 1684.2
- ULTIMATE BRACING CAPACITY 6kN/m
- REFER TO PLAN FOR LOCATION AND LENGTH OF BRACING



NOTE: NOGGINGS HAVE BEEN OMITTED FOR CLARITY

TYPICAL WALL BRACING DETAIL – TYPE B

NOTE: IN ACCORDANCE WITH REQUIREMENTS OF AS1684 NATIONAL FRAMING CODE.



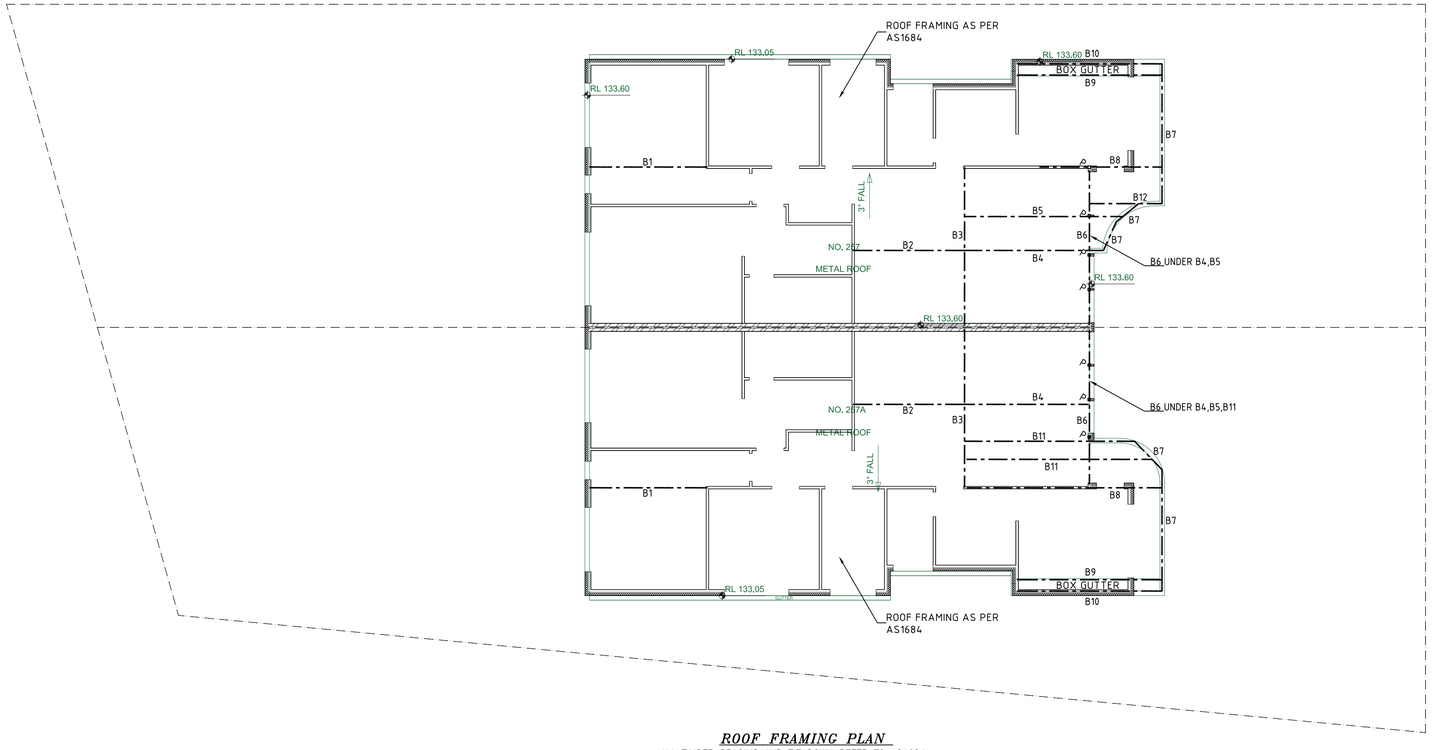
HARDBOARD BRACING DETAIL – TYPE C

1. FIX HARDBOARD PANELS WITH 30X28MM DIA MIN. OR EQUIVALENT GALVANIZED FLATHEAD NAILS @ 40MM CENTRES LONG TOP AND BOTTOM PLATES,150MM CENTRES ALONG VERTICAL EDGES AND NOGGIES
2. NAILS SHALL BE LOCATED A MIN.10MM FROM VERTICAL EDGES AND 15MM FROM THE TOP AND BOTTOM EDGES.
3. MIN. HARDBOARD THICKNESS=6MM
4. HARDBOARD SHALL COMPLY WITH AS/NZS1859.4

client	date	issue	amendment	checked	date	issue	amendment	checked	project
	16.09.2026	A	ISSUED FOR CC	JZ					257 MARSDEN ROAD, CARLINGFORD

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PARRAMATTA NSW 2124
MBCCONSULTING@LIVE.COM
MOBILE:0432546227

drawing title: BRACING LAYOUT PLAN
date 16/09/2026
job no: 2025466
scale 1:100,1:20
designed: C.Z.(BE,MIE 3928680)
drafted: J.Z.
drawing no. S05



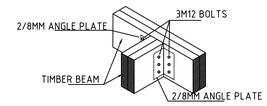
ROOF FRAMING PLAN

ALL TIMBER BRACING AND TIE DOWN REFER TO AS1684
REFER TO TIMBER MANUFACTURER'S SEPCIFICATION AND DETAILS

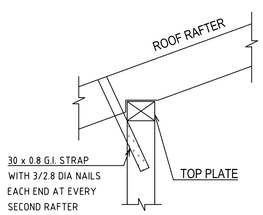
BEAM SCHEDULE

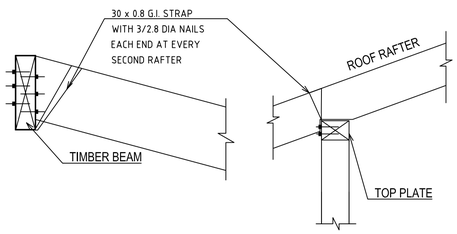
MARK No.	SIZE
B1	2/200x42 SMARTLVL 15
B2	2/200x42 SMARTLVL 15
B3	2/300x42 SMARTLVL 15
B4	2/200x42 SMARTLVL 15
B5	2/200x42 SMARTLVL 15
B6	2/200x42 SMARTLVL 15
B7	300x42 SMARTLVL 15
B8	2/200x42 SMARTLVL 15
B9	2/200x42 SMARTLVL 15
B10	2/200x42 SMARTLVL 15
B11	2/200x58 SMARTLVL 15
B12	2/200x42 SMARTLVL 15
P	90x90 HW POST

ALL EXTERNAL STEEL BEAM TO BE H.D.GALV.
SURFACE PREPARATION & FINISH TO COMPLY WITH AS/NZS 2312/1994
ALL EXTERNAL TIMBER BEAM TO BE TREATED AS H3

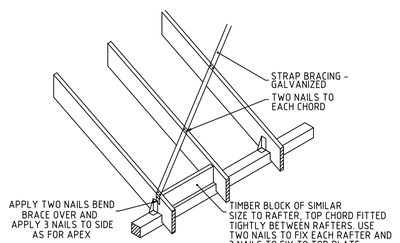


TYPICAL TIMBER BEAM CONNECTION DETAILS

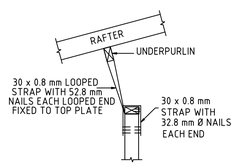




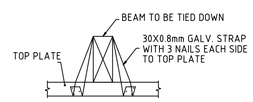
TYPICAL RAFTER TIE-DOWN DETAIL



ROOF BRACING FIXING TO RAFTERS AT EAVES
SCALE 1:5



UNDERPURLIN TO TOP PLATE TIE-DOWN DETAIL
TOP PLATE TO STUD WORK TIE-DOWN
SCALE 1:5



TIE DOWN DETAIL (TYPICAL)
SCALE 1:5

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drafted: J.Z.
drawing no. S06